METRIC

A-A-59003/10
31 May 1995
Superseding
MIL-A-21577/10
31 July 1968

COMMERCIAL ITEM DESCRIPTION

AMPLIFIER, AUDIO FREQUENCY - EMERGENCY

REPORT SYSTEM

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-A-21577/10 which is canceled.

1. **Scope**. This commercial item description covers the detailed requirements for an amplifier used to amplify audio frequency signals to drive a group of loudspeakers in an emergency report system onboard a submarine.

2. **Salient characteristics**.

2.1 **General**. The amplifier shall be in accordance with the applicable paragraphs of A-A-59003 and as specified herein. Whenever a requirement of A-A-59003 conflicts with a requirement of this specification, the requirement of this specification shall govern.

2.1.1 **Equipment**. The equipment complement shall consist of two amplifiers housed in one enclosure.

2.2 **Equipment function**. The function of the equipment described herein shall be to provide the amplification of voice signals required for emergency report system circuit 4MC.

2.2.1 **Circuit 4MC function**. The function of the emergency report system, circuit 4MC is to provide one-way transmission of communications of an emergency nature from stations located throughout the ship to the ship's control center and to the maneuvering area.

2.2.2 **Other components**. Other components used in circuit 4MC but not covered by this specification are as specified in 3.2.2.1 and 3.2.2.2.

2.2.2.1 **Emergency report switches**. Emergency report switches, type AN/SIA-718A shall be used to transfer sound powered telephones from their normal circuit to circuit 4MC at input stations. Operation of any of these switches shall energize the amplifier.

2.2.2.2 **Loudspeakers**. Type LS-305 and LS-387 loudspeakers in accordance with A-A-59002 shall be used at all locations using loudspeakers.

2.3 **Enclosure**. The enclosure shall be of drip-proof construction designed for bulkhead mounting. Amplifiers shall be drawout-type mounted on guide rails

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and connected by male and female connectors. Provision shall be made for front servicing of all components. Dowels and matching bushings shall be provided for support and alignment of units.

2.4 **Size and weight**. Maximum overall dimensions of the enclosure shall be 305 mm in width, 381 mm in height, and 305 mm in depth. Minimum practical dimensions consistent with design requirements will be acceptable. The weight of the unit shall not exceed 18.14 kg.

2.5 **Semiconductors**. Semiconductors circuits shall be used exclusively.

2.6 **Relays**. Relays may be hermetically sealed.

2.7 **Output power**. The rated output power of each amplifier shall be 10.0 watts minimum.

2.8 **Input facilities**. Terminal blocks shall be provided to accommodate four input circuits. Individual emergency report station input circuits shall be paralleled external to the amplifier enclosure to limit the number of input circuit terminals required.

2.9 **Power supply**. Normal power input shall be from 115 volt, 60-cycle, single-phase ship's service. A nickel cadmium battery shall be provided as an emergency supply in the event of failure of the normal supply. A box for mounting the battery exterior to the amplifier enclosure shall be provided.

2.10 **Active circuit**. One amplifier shall be in the active circuit. The other amplifier shall be considered as a spare. The spare amplifier shall be used as a replacement module in the event of failure of the active module.

2.11 **Test facilities**. Test facilities shall be provided to test the amplifier in the spare position. A voltmeter shall be provided to measure the output of the amplifier. A test jack shall be provided to permit use of a microphone to insert a test signal to measure amplifier output.

3. **Quality assurance**. Sampling, examination, and tests shall be in accordance with A-A-59003.

4. **Packaging**. Packaging shall be in accordance with A-A-59003.

5. **Notes**.

5.1 **Intended use**. The amplifier, audio frequency is intended for use in emergency report system, circuit 4MC on board submarines.

5.2 **Notes**. The notes specified in A-A-59003 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 6320-N056)

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